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Zhu

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(54) **METHOD FOR ENHANCING INTERIOR
DISPLAY READABILITY IN MOTORIZED
OBJECTS AND SYSTEM THEREFOR**

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B60J 3/04 (2006.01)

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B60K 35/22 (2024.01)

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CPC . **B60J 3/06** (2013.01); **B60J 3/04** (2013.01);
B60K 35/00 (2013.01); **B60K 35/22** (2024.01);
B60K 35/415 (2024.01); **B60Y 2200/11**
(2013.01)

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3/00–06; **B60Y 2200/00–92**; **G02B**
27/00–648; **G02B 2027/0105–0198**

See application file for complete search history.

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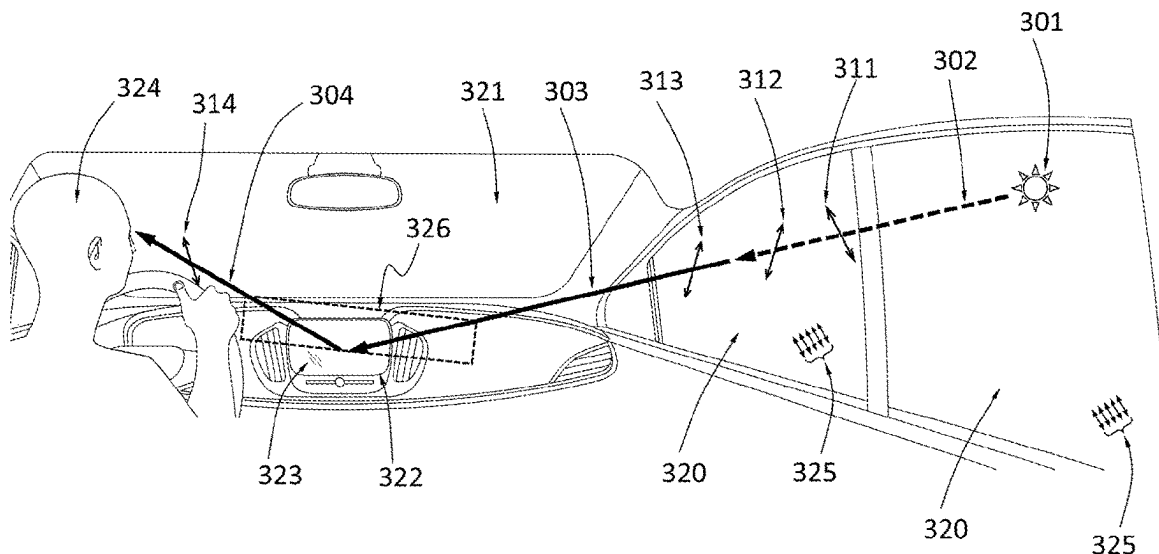
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Manni Li

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ABSTRACT

A method for improving readability of a display inside a motorized object comprises applying a polarization means on a side window of a motorized object, wherein an ambient incident light ray comes through the side window and is reflected by a surface of a display inside the motorized object to form a reflected light ray, and substantially filtering out display incident light S-polarization and allowing display incident light P-polarization to pass through. The polarization means has a linear polarization reflecting or absorbing direction that is substantially parallel to the plane of the surface of the display. A display readability enhancing system comprises the polarization means. When ambient light passes through the window and shines onto the interior display, the dominant light polarization is p-polarization for the interior display with minimum surface reflectivity upon reflecting from the interior display, resulting in good display visibility under strong ambient light conditions.

18 Claims, 12 Drawing Sheets



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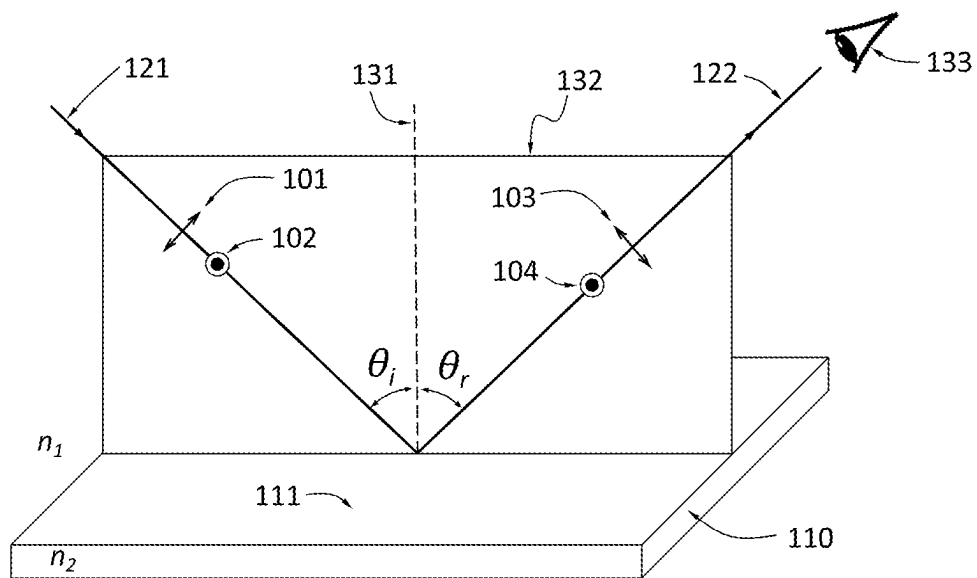


FIG. 1

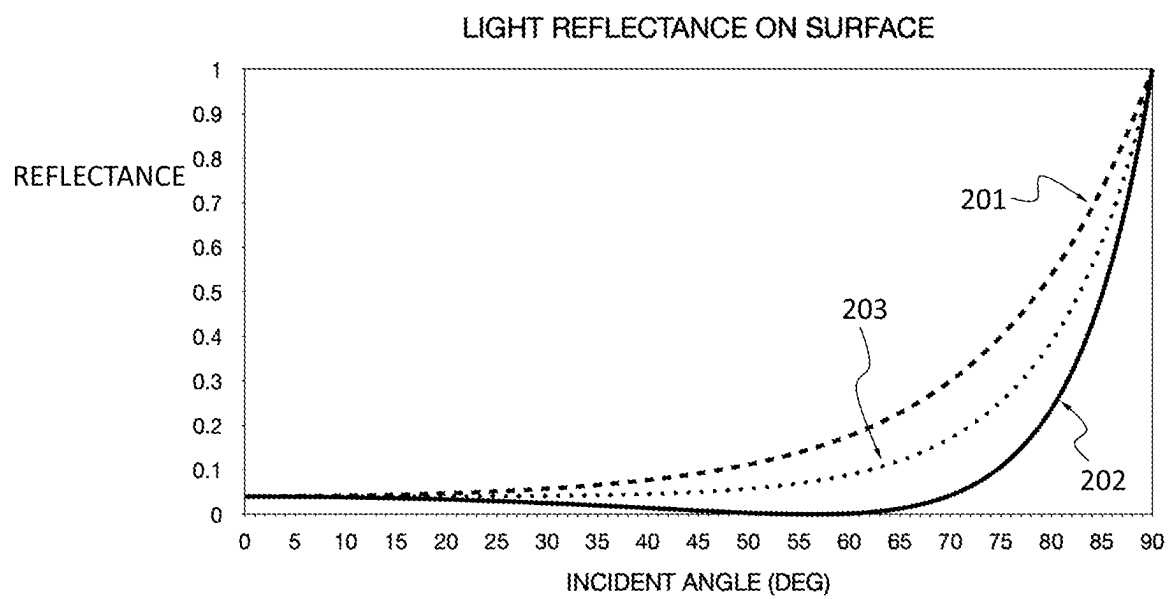


FIG. 2

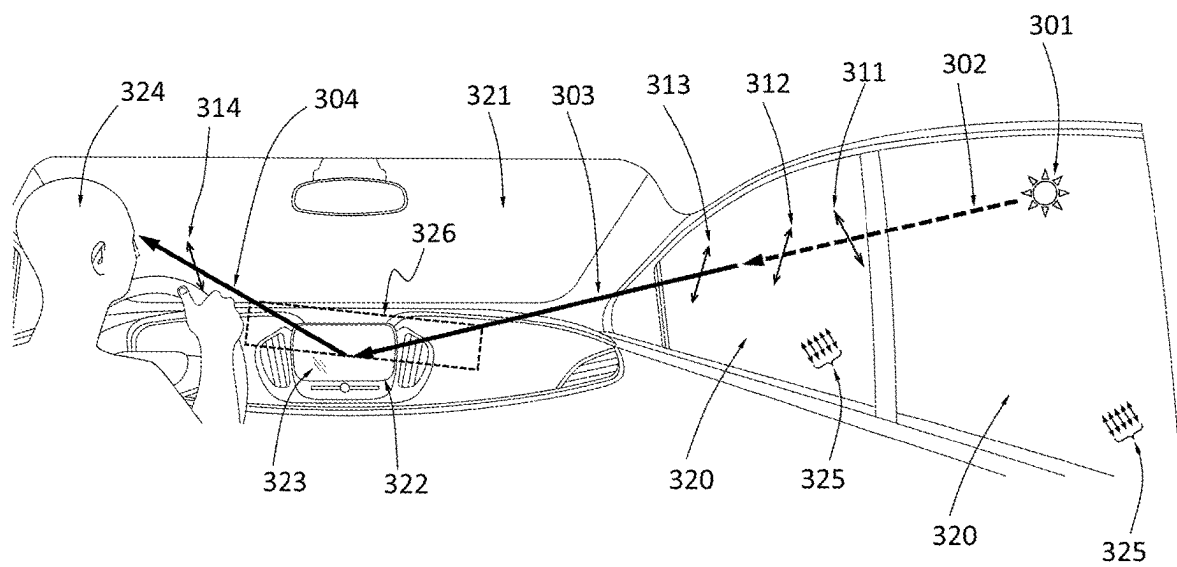


FIG. 3A

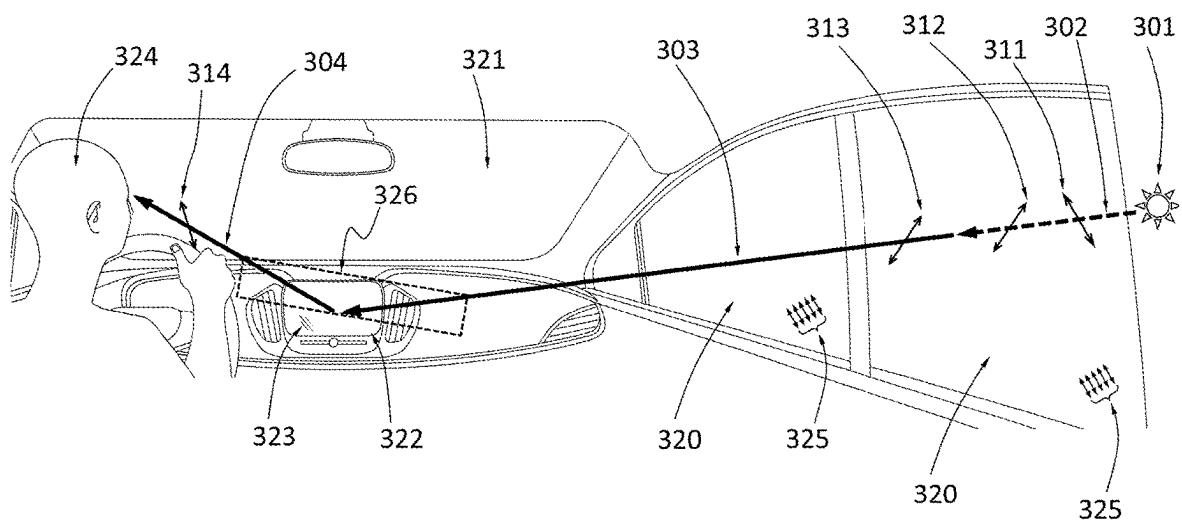


FIG. 3B

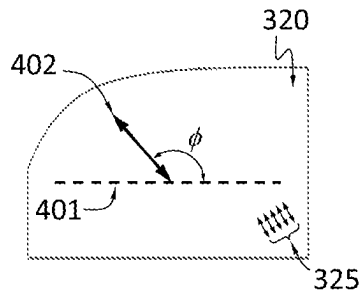


FIG. 4A

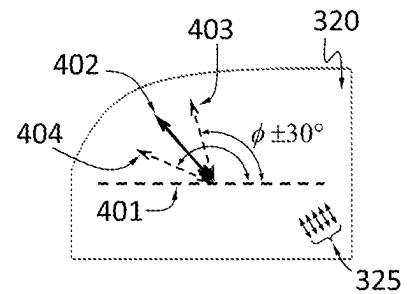


FIG. 4B

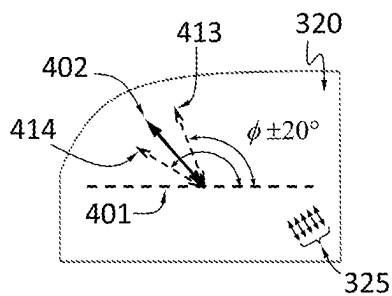


FIG. 4C

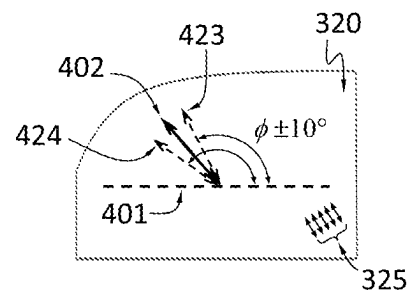


FIG. 4D

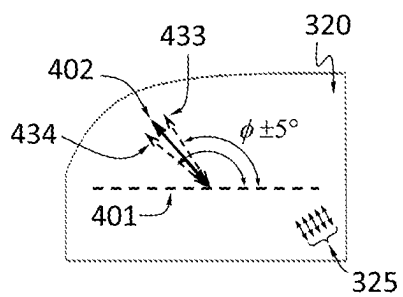


FIG. 4E

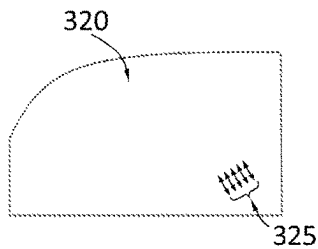


FIG. 5A

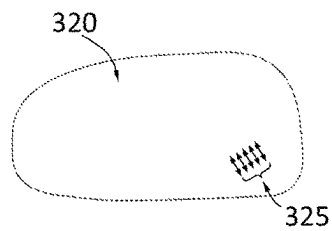


FIG. 5B

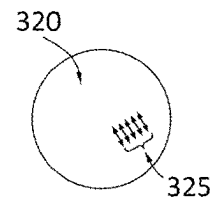


FIG. 5C

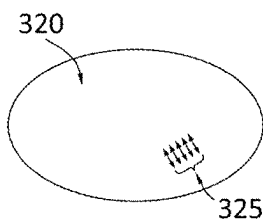


FIG. 5D

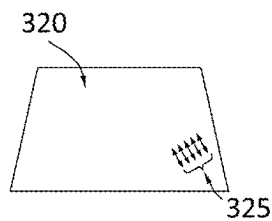


FIG. 5E

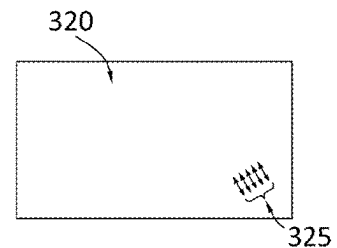


FIG. 5F

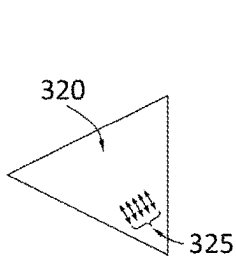


FIG. 5G

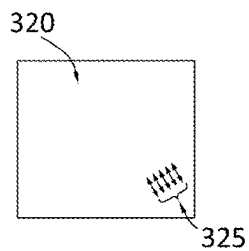


FIG. 5H

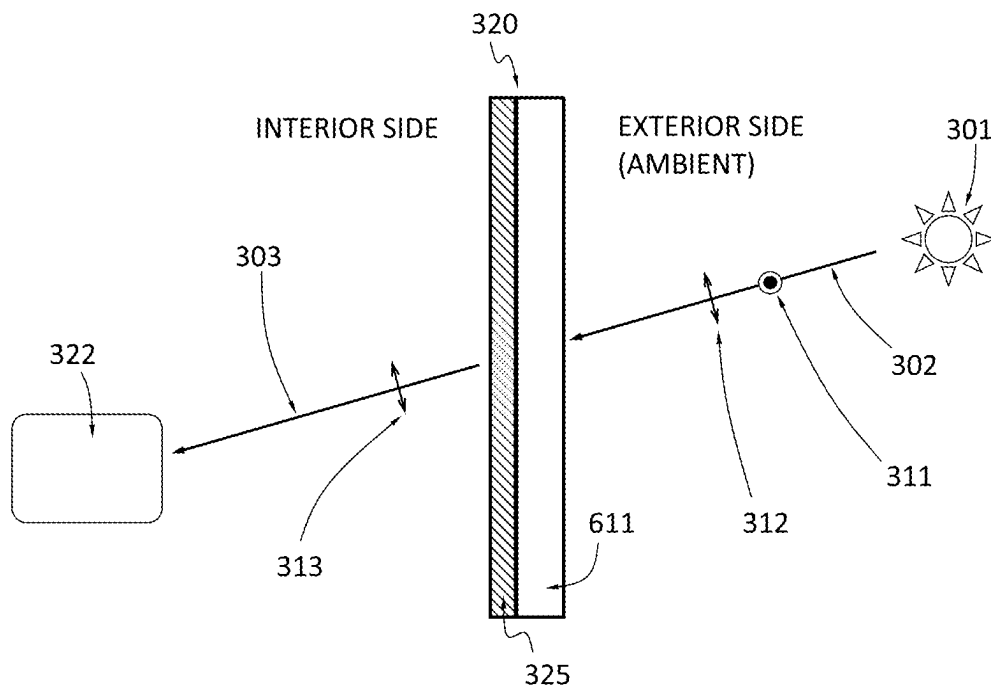


FIG. 6

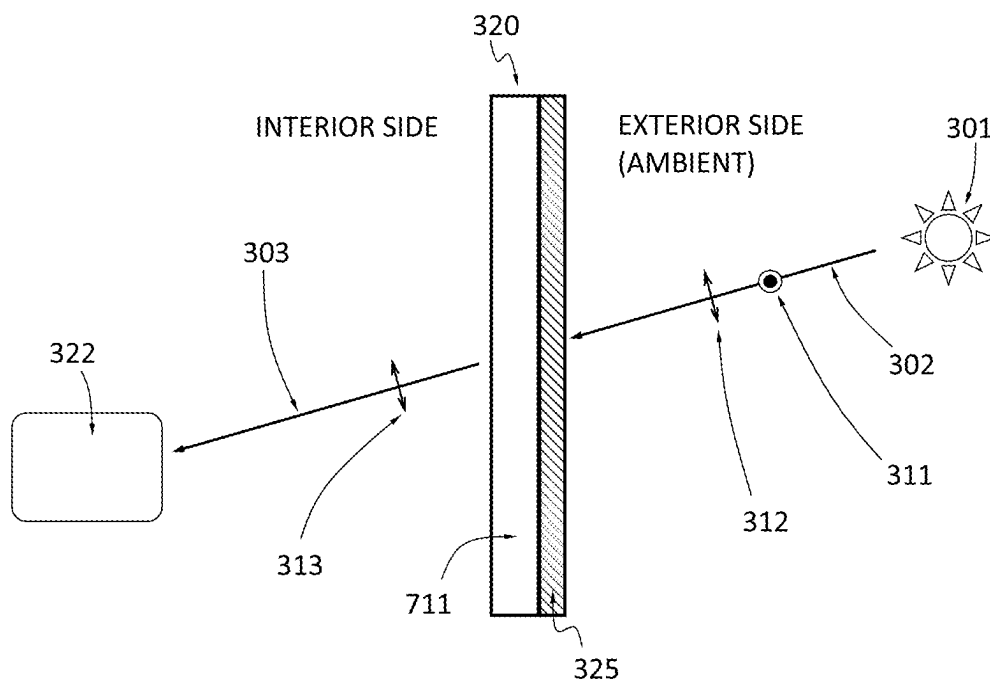


FIG. 7

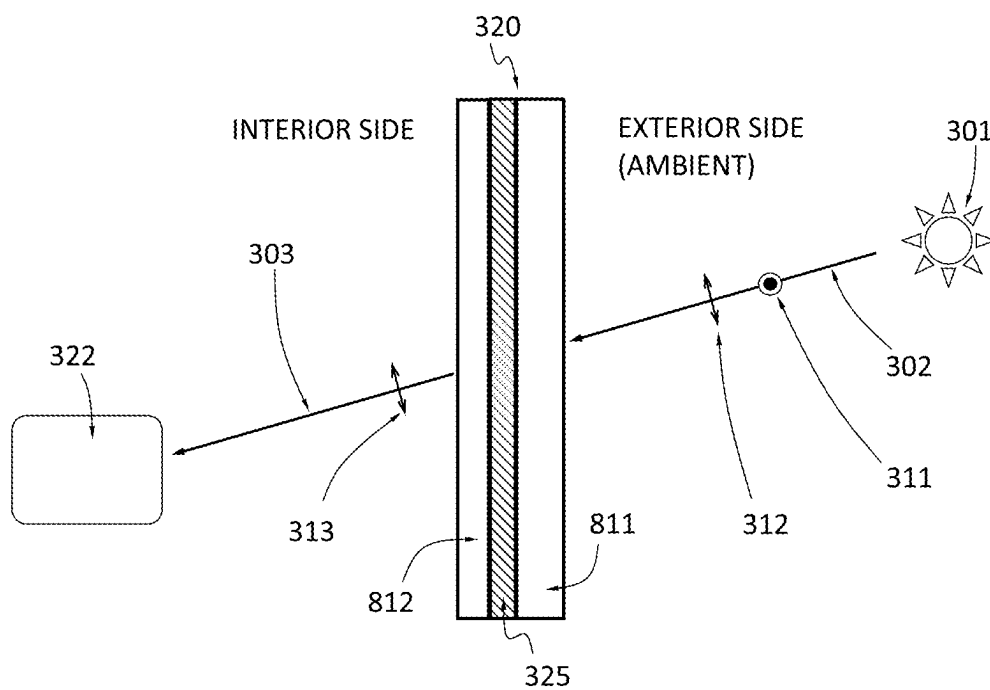


FIG. 8

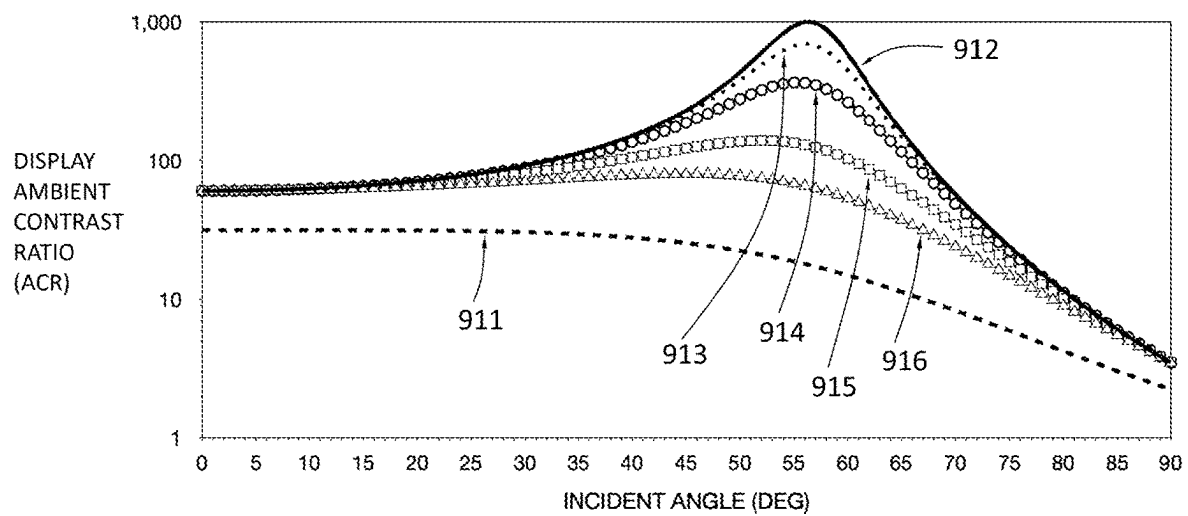


FIG. 9A

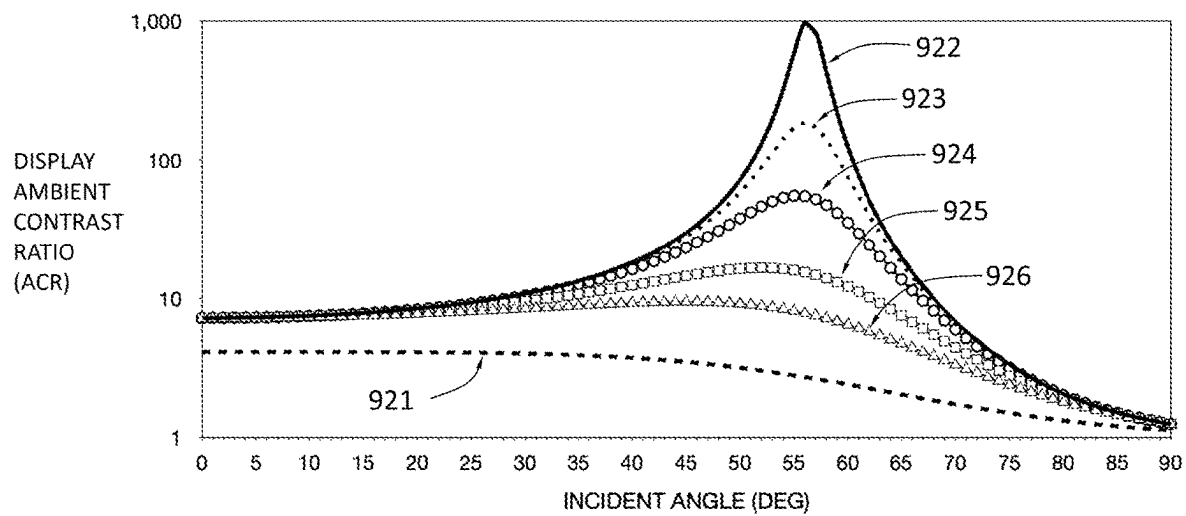


FIG. 9B

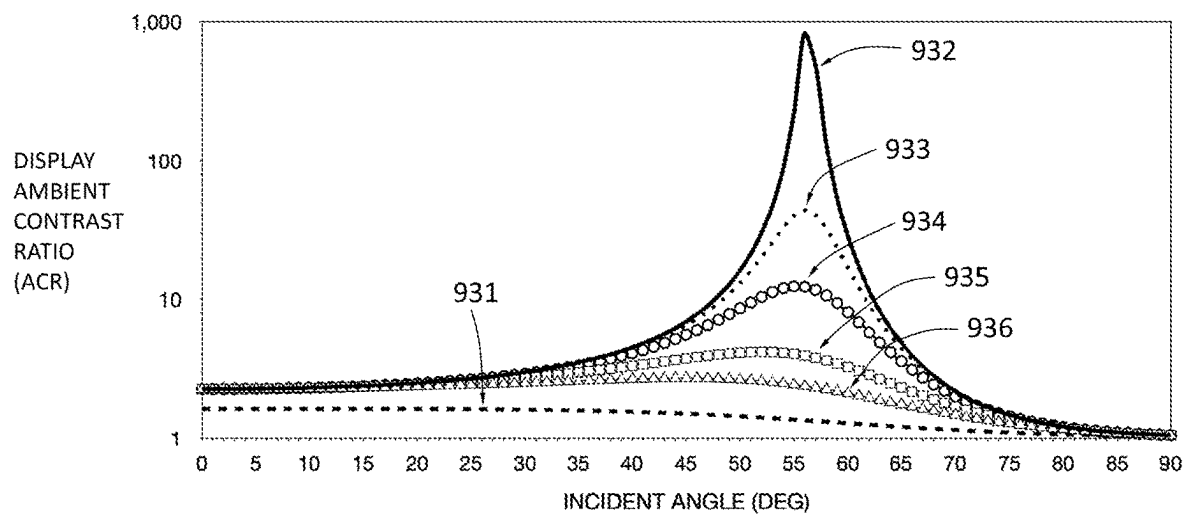


FIG. 9C

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METHOD FOR ENHANCING INTERIOR DISPLAY READABILITY IN MOTORIZED OBJECTS AND SYSTEM THEREFOR

TECHNICAL FIELD

The present invention relates to method and system to enhance the readability of the interior display in a motorized object by applying on the side window with light polarization means.

BACKGROUND OF INVENTION

Interior displays are critical components for many motorized objects, such as vehicles, boats, planes, etc. In some cases, there are more than one display in the motorized object. For example, there are usually two displays in a vehicle: one is in the dashboard in front of the driver, showing the driving condition information; the other is located in the front and between the driver and passenger, functioning as the vehicle infotainment display. A major issue with the interior display is that the readability of the display can be easily impacted by the ambient light conditions.

For vehicle drivers and passengers, the display readability at different ambient conditions is important as they need to read information from the display or interact with the display. A strong sunlight reflection from the display could easily wash out the display image so that the driver and passenger could not see the display information clearly. Such bad display readability under ambient light conditions may cause severe driving safety issues. As the interior display becomes more important, and sometimes the key feature, in the operation of the motorized objects, the interior display readability issue needs to be directly addressed, however, it has been largely ignored by the industry.

SUMMARY OF INVENTION

The present invention solves the interior display readability issue in the motorized object under different ambient light conditions and provides a method for enhancing the interior display readability in a motorized object.

The present invention provides a method for improving the readability of a display inside a motorized object, comprising applying a polarization means on a side window of a motorized object, wherein an ambient light propagates through the side window to become a display incident light, the display incident light is reflected by a surface of a display inside the motorized object to form a reflected light, and the display incident light and the reflected light form a display incident plane, and substantially filtering out S-polarization relative to the display incident plane from the ambient light and allowing P-polarization relative to the display incident plane from the ambient light to pass through the side window by the polarization means, wherein the polarization means has a linear polarization reflecting or absorbing direction that is substantially parallel to a plane of the surface of the display; the polarization means is on the side window that is on a passenger side, a driver side, or both the passenger and driver sides of the motorized object; and the side window is a front side window, a rear side window, or both front and rear side windows.

In the present invention, the polarization means may be passively applied on the side window or switchable on the side window.

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In the present invention, the side window may be of a round, oval, trapezoid, rectangle, square, triangle, or irregular shape.

In the present invention, the linear polarization reflecting or absorbing direction of the polarization means may have no more than 30 degree deviation from being parallel to the plane of the surface of the display; preferably, no more than 20 degree deviation from being parallel to the plane of the surface of the display; more preferably, no more than 10 degree deviation from being parallel to the plane of the surface of the display; and most preferably, no more than 5 degree deviation from being parallel to the plane of the surface of the display.

In the present invention, the polarization means may filter out at least 75% of the S-polarization from the ambient light; preferably, at least 88% of the S-polarization from the ambient light; more preferably, at least 97% of the S-polarization from the ambient light; and most preferably, at least 99% of the S-polarization from the ambient light.

The present invention further provides a display readability enhancing system for use in the method of the present invention which comprises the polarization means. The polarization means in the present invention has the linear polarization reflecting or absorbing direction that is substantially parallel to the plane of the surface of the display; and the polarization means is on the side window that is on the passenger side, the driver side, or both the passenger and driver sides of the motorized object.

In the display readability enhancing system of the present invention, the polarization means may be located on an interior side of the side window or on an exterior side of the side window, or sandwiched between two layers of the side window.

In the display readability enhancing system of the present invention, the polarization means of the display readability enhancing system may be a passive means or switchable.

In the display readability enhancing system of the present invention, the polarization means may filter out at least 75% of the S-polarization from the ambient light.

The present invention improves the vehicle's display readability by implementing a polarization means on the vehicle's side windows so that the side windows filter out the ambient incident light's S-polarization, while only allowing P-polarization to pass through.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows the reflection of the ambient incident light by the surface of the display and propagates towards eye viewing direction as an issue to be addressed in the present invention.

FIG. 2 shows the display surface light reflectance of incident light with P-polarization, S-polarization, or both, at different incident angles according to the Fresnel Equation in the present invention. Dashed curve **201** is the light reflectance of the S-polarization R_s . The solid curve **202** is the light reflectance of the P-polarization. The dotted curve **203** is the total light reflectance R_{TOTAL} .

FIGS. 3A and 3B show the method and system for improving display readability of the present invention, wherein FIG. 3A shows the light propagation and polarization change effect of the present invention implemented in one side window of a vehicle, and FIG. 3B shows the light propagation and polarization change effect of the present invention implemented in a further side window of a vehicle.

FIGS. 4A to 4E show side views of the angle of linear polarization reflecting or absorbing direction of the polar-

ization means on the side window in the present invention, where FIG. 4A shows the preferred polarization reflecting or absorbing direction parallel to the display surface plane; FIG. 4B shows the direction with a deviation of 30° from the preferred direction; FIG. 4C shows the direction with a deviation of 20° from the preferred direction; FIG. 4D shows the direction with a deviation of 10° from the preferred direction; FIG. 4E shows the direction with a deviation of 5° from the preferred direction.

FIGS. 5A to 5H show side views of the side windows in different shapes where the method and system for improving display readability of the present invention apply, wherein FIG. 5A shows an irregular-shaped window with straight horizontal and vertical sides, FIG. 5B shows an irregular-shaped window with round edges, FIG. 5C shows a circular window, FIG. 5D shows an oval window, FIG. 5E shows a trapezoidal window, FIG. 5F shows a rectangular window, FIG. 5G shows a triangular window, and FIG. 5H shows a square-shaped window.

FIG. 6 is a cross-sectional view showing the first embodiment of the present invention where the polarization means is located at the interior side of the side window glass.

FIG. 7 is the cross-sectional view showing the second embodiment of the present invention where the polarization means is located at the exterior side (ambient lighting side) of the side window glass.

FIG. 8 is a cross-sectional view showing the third embodiment of the present invention where the polarization means is located between the interior glass and the exterior glass of the side window glass.

FIGS. 9A to 9C are graphs showing the change in display ambient contrast ratio (ACR) in relation to the incident angle (unit: DEG=degree, also depicted by the symbol ° in the present application) according to Examples 1-3 in the present invention, where FIG. 9A shows the dependency of the display ACR on the light incident angle in the case of the ambient illuminance 2000 lux in Example 1; FIG. 9B shows the dependency of the display ACR on the light incident angle in the case of the ambient illuminance 20000 lux in Example 2; and FIG. 9C shows the dependency of the display ACR on the light incident angle in the case of the ambient illuminance 100000 lux in Example 3.

DETAILED DESCRIPTION OF INVENTION

The present invention is described in detail in connection with the drawings. Any one of ordinary skill in the art may modify or change the present invention as described below without departing from the scope of protection for the present invention.

The method and system for improving display readability of the present invention is useful in motorized objects, including but not limited to, passenger or unmanned vehicles, boats, aircraft, watercrafts, truck, train, etc., having side windows typically made of transparent glass or plastic and display panel in the interior.

As shown in FIG. 1, when an ambient light shines onto an object's surface, there is generally a light reflection from the object's surface. Such ambient light surface reflection can add noise to a display image quality and affect the display's readability. Ambient incident light 121 propagates from an ambient medium having refractive index n_1 to display 110 having refractive index n_2 . Ambient incident light 121 is reflected by surface 111 of display 110 and propagates towards the viewer's eye 133. Incident light 121, a surface normal line 131 that is perpendicular to display surface 111, and reflected light 122 form light incident plane 132. Inci-

dent light 121 comprises both P-polarization 101 and S-polarization 102. P-polarization 101 is parallel to incident plane 132. S-polarization 102 is perpendicular to incident plane 132 and parallel to surface 111 of display 110. Reflected light 122 also comprises both P-polarization 103 and S-polarization 104. Angle θ_i is light incident angle, and angle θ_r is light reflecting angle.

In the present invention, unless otherwise noted, "P-polarization" is referring to the display incident light P-polarization that is parallel to the display incident plane, and "S-polarization" is referring to the display incident light S-polarization that is perpendicular to the display incident plane and parallel to the display surface.

The display may have a flat surface or a curved surface. A display center surface tangential plane may be defined as follows. For a curved display surface, the display center surface tangential plane is the plane tangential to the display curved surface center. For a display with a flat surface, the display center surface tangential plane is the same plane as the display surface. In the present invention, unless otherwise noted, "display surface" or "surface of display" is referring to the display center surface tangential plane.

The light reflection from the display surface may be quantified by the following Fresnel Equations:

$$R_P = \frac{\left| n_1 \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_i \right)^2} - n_2 \cos \theta_i \right|^2}{\left| n_1 \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_i \right)^2} + n_2 \cos \theta_i \right|^2}, \quad (1)$$

$$R_S = \frac{\left| n_1 \cos \theta_i - n_2 \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_i \right)^2} \right|^2}{\left| n_1 \cos \theta_i + n_2 \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_i \right)^2} \right|^2}, \text{ and} \quad (2)$$

$$R_{TOTAL} = \frac{R_P + R_S}{2}, \quad (3)$$

where R_P is the light reflectance of P-polarization, R_S is the light reflectance of S-polarization, and R_{TOTAL} is the total light reflectance when the incident light has both P-polarization and S-polarization. If the incident light has only one polarization, such as P-polarization, then the total light reflectance R_{TOTAL} equals to R_P . Further, n_1 is the light refractive index of the ambient medium, n_2 is the light refractive index of the substrate of the display 110 in FIG. 1, and θ_i is the light incident angle.

As shown in FIG. 2, the display surface light reflectance at different incident angles changes according to the Fresnel Equation. S-polarization reflectance R_S in dashed curve 201 keeps on increasing as the incident angle increases. On the other hand, P-polarization reflectance R_P in solid curve 202 gradually decreases from 0° incident angle to around 56° incident angle. At around 56° incident angle, P-polarization reflectance R_P reaches 0. The angle is called the Brewster Angle. When the incident angle increases further from the Brewster Angle, R_P increases quickly. If the incident light comprises an equal intensity of both P-polarization and S-polarization, then the total reflectance R_{TOTAL} in dotted curve 203 monotonously increases as the incident angle increases.

When a display is exposed in ambient light conditions, the display surface's light reflectance adds noise to the display image, which causes the display readability issue. As shown in FIG. 2, when the incident angle increases, total reflectance R_{TOTAL} in dotted curve 203 increases more, resulting in even

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poorer display readability at large viewing angles. S-polarization's reflectance R_S in dashed curve **201** is the major contributing factor causing total reflectance R_{TOTAL} in dotted curve **203** increase when the incident angle increases. The present invention provides the method and system to filter out S-polarization from the ambient incident light while retaining the P-polarization, therefore, keeping total surface reflectance R_{TOTAL} in the reflected light relatively low, which equals to R_P (less than about 4%) from 0° to 70° incident angle as shown in solid curve **202** in FIG. 2. Such a low surface reflectance significantly reduces the ambient light noise added onto the display image when the viewer sees the display; therefore, the display readability can improve greatly.

The method for improving readability of a display inside a motorized object in the present invention applies a polarization means to substantially filter out S-polarization in the ambient incident light and allow P-polarization in the ambient incident light to pass through the side window by the polarization means. At least about 65% of the S-polarization is filtered out, for example, at least 70%, and preferably, at least 75% of the S-polarization is filtered out.

In the present invention, the display's native contrast ratio is defined by the equation below:

$$\text{DISPLAY NATIVE CONTRAST RATIO} = \frac{L_{BRIGHT}}{L_{DARK}}, \quad (4)$$

where L_{BRIGHT} is the display luminance at the display bright state, while L_{DARK} is the display luminance at display dark state. If the display is used in an ambient lighting condition, then the display's readability can be presented using the display ambient contrast ratio (ACR), as defined by the equation below:

$$\text{DISPLAY AMBIENT CONTRAST RATIO (ACR)} = \frac{L_{BRIGHT} + \frac{I_{AMBIENT}}{\pi} \cdot R_{SURFACE}}{L_{DARK} + \frac{I_{AMBIENT}}{\pi} \cdot R_{SURFACE}}, \quad (5)$$

where L_{BRIGHT} is the display luminance at the display bright state, while L_{DARK} is the display luminance at display dark state. $I_{AMBIENT}$ is the illuminance of the ambient lighting condition. A typical overcast day illuminance is in the range of 1000 to 4000 Lux. A typical full daylight illuminance is in the range of 8000 to 30000 Lux. A typical direct sunlight illuminance is in the range of 80000 to 200000 Lux. $R_{SURFACE}$ is the display surface reflectance, and π is a constant ($\pi \approx 3.1415926$).

As shown in FIG. 3A, the light propagation and polarization change effect is shown as the method and system of the present invention are implemented in one side window of a vehicle. In the present invention, the side window may be any side windows at the driver side or the passenger side, at the front row or the rear rows. The vehicle comprises a side window **320**, a front windshield window **321**, and a main display **322** with a display surface **323**. The ambient light **302** from the ambient light source **301** (such as sunlight) passes through the side window **320**. After passing through the side window **320**, the ambient light **302** becomes the incident light **303**. Then the incident light **303** is reflected by the display surface **323**. Upon reflection from the display

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surface **323**, the reflected light **304** reaches the viewer **324**. The incident light **303** and the reflected light **304** form an incident plane **326**. The ambient light **302** is an unpolarized light, which has S-polarization **311** and P-polarization **312**; the S-polarization **311** is parallel to the display surface **323**, and the P-polarization **312** is parallel to the incident plane **326**. The side window **320** has a polarization means **325** that absorbs or reflects the S-polarization **311** of the ambient light **302**, but allows P-polarization **312** to pass through. As the result, the incident light **303** has P-polarization **313** only, which is parallel to the incident plane **326**. Similarly, the reflected light **304** also only has the P-polarization **314**, which is parallel to the incident plane **326**.

As shown in FIG. 3B, the ambient light may pass through other side windows, such as the rear row side windows. Similarly, to those shown in FIG. 3A, the ambient light **302** passes through the second-row passenger side window **320**, then the incident light **303** is reflected by the display surface **323** of the display **322** and reaches the viewer **324**.

In FIGS. 3A and 3B, the side window **320** has the polarization means **325** with filtering function that absorbs or reflects the S-polarization **311** of the ambient light **302** so that the incident light **303** has P-polarization only. Such a P-polarization-only incident light **303** has low display surface reflectance R_P as shown in FIG. 2.

As shown in FIGS. 4A to 4E, the side window **320**'s linear polarization means have a linear polarization reflecting or absorbing direction, which is preferably parallel to the display surface plane. With such a preferred linear polarization reflecting or absorbing direction, the display's incident light only has P-polarization, no S-polarization. Therefore, the display reflectance is low, and the display readability is good at different ambient conditions.

The side windows' linear polarization reflecting or absorbing direction may deviate from the preferred direction. The side window's linear polarization reflecting or absorbing direction forms an angle with respect to the display surface plane: in the present invention, the angle is preferably no more than 30° , preferably no more than 20° , more preferably no more than 10° , and most preferably no more than 5° .

FIG. 4A is the side view of the side window **320** having a polarization means which has a preferred linear polarization reflecting or absorbing direction **402** that is parallel to the display surface plane. The side window **320** absorbs or filters out the linear polarization direction **402** of the incoming ambient light. The linear polarization direction **402** is parallel to the display surface plane and it forms an orientation angle ϕ with respect to the horizontal reference direction **401**. Therefore, the linear polarization direction **402** is exactly the S-polarization for the incident light onto the display surface. As the linear polarization direction **402** is the reflecting or absorbing direction of the polarizer means of the side window **320**, when the ambient light shines onto the side window, only P-polarization can pass through the side window. Such P-polarization only incident light onto display has low surface reflectance and provides good ambient light readability.

FIG. 4B shows the side view of the side window **320** having a polarization means which has a linear polarization reflecting or absorbing direction deviating from the preferred linear polarization reflecting or absorbing direction **402** by $\pm 30^\circ$. The dashed arrow directions **403** and **404** show the deviation range boundary. The preferred linear polarization reflecting or absorbing direction **402** is parallel to the display surface plane.

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FIG. 4C shows the side view of the side window 320 having a polarization means which has a linear polarization reflecting or absorbing direction deviating from the preferred linear polarization reflecting or absorbing direction 402 by $\pm 20^\circ$. The dashed arrow directions 413 and 414 show the deviation range boundary. The preferred linear polarization reflecting or absorbing direction 402 is parallel to the display surface plane.

FIG. 4D shows the side view of the side window 320 having a polarization means which has a linear polarization reflecting or absorbing direction deviating from the preferred linear polarization reflecting or absorbing direction 402 by $\pm 10^\circ$. The dashed arrow directions 423 and 424 show the deviation range boundary. The preferred linear polarization reflecting or absorbing direction 402 is parallel to the display surface plane.

FIG. 4E shows the side view of the side window 320 having a polarization means which has a linear polarization reflecting or absorbing direction deviating from the preferred linear polarization reflecting or absorbing direction 402 by $\pm 5^\circ$. The dashed arrow directions 433 and 434 show the deviation range boundary. The preferred linear polarization reflecting or absorbing direction 402 is parallel to the display surface plane.

In the present invention, the polarization means on the side windows may be applied in different shapes of side windows. For examples, FIGS. 5A to 5H show different shapes side windows including round, oval, trapezoid, rectangle, square, triangle, or irregular shapes.

In the present invention, to implement the side window polarization means with the filtering function, the linear polarization filtering means can be located at different interfaces of the side window stack.

FIG. 6 shows the cross-sectional view of the first embodiment of the present invention where the linear polarization means is located at the interior side of the side window. The side window 320 comprises a transparent layer 611 and a linear polarization means 325. The linear polarization means 325 is located at the interior side, and the transparent layer 611 is located at the exterior side (ambient light source side). The ambient light 302 from the ambient light source 301 has both P-polarization 312 and S-polarization 311. When the ambient light 302 passes through the side window 320, the linear polarization means 325 filters out the S-polarization 311 while allowing the P-polarization 312 to pass through. As a result, the incident light 303 onto the display 322 has P-polarization 313 only.

FIG. 7 shows the cross-sectional view of the second embodiment of the present invention where the linear polarization means is located at the exterior side (ambient lighting side) of the side window glass. The side window 320 comprises a transparent layer 711 and a linear polarization means 325. The linear polarization means 325 is located at the exterior side (ambient light source side), and the transparent layer 712 is located at the interior side. The ambient light 302 from the ambient light source 301 has both P-polarization 312 and S-polarization 311. When the ambient light 302 passes through the side window 320, the linear polarization means 325 filters out the S-polarization 311 while allowing the P-polarization 312 to pass through. As a result, the incident light 303 onto the display 322 has P-polarization 313 only.

FIG. 8 shows the cross-sectional view of the third embodiment of the present invention where the linear polarization means is located in between the interior glass and the exterior glass of the side window glass. The side window 320 comprises a first transparent layer 811, a second trans-

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parent layer 812, and a linear polarization means 325. The linear polarization means 325 is sandwiched between the first transparent layer 811 and the second transparent glass layer 812. The ambient light 302 from the ambient light source 301 has both P-polarization 312 and S-polarization 311. When the ambient light 302 passes through the side window 320, the linear polarization means 325 filters out the S-polarization 311 while allowing the P-polarization 312 to pass through. As a result, the incident light 303 onto the display 322 has P-polarization 313 only.

In the present invention, the polarization means for reflecting or absorbing the S-polarization may be passively applied or switchable as to the polarization properties.

The first method is to have a linear polarizer layer applied onto the side window glass surface or sandwiched between the side window glasses. The linear polarizer layer may be produced by dyeing a polyvinyl alcohol (PVA) based film with a dichroic dye such as iodine and then stretching it in an aqueous boric acid solution. When the incident light passes through such a linear polarizer, one linear polarization is absorbed by the polarizer while the orthogonal linear polarization can pass through. The passing through linear polarization is determined by the polarizer PVA layer stretching direction. The manufacturing of linear polarization means is known in the art, such as a linear polarizer film layer, referring to U.S. Pat. No. 7,110,177B2 and U.S. Patent Application Publication No. 20030001988A1 for detailed manufacturing material and process to make the linear polarizer film, which are incorporated herein by reference.

A second method is to have a switchable polarizing means sandwiched between the side window glass. Such switchable polarizer means can be electrically controlled and change the passing-through linear polarization orientation. When the linear polarization means in the side window is an adaptive means, such as an electronically controlled guest-host liquid crystal device using dichromatic or dichroic dye material, it is known in the art how to manufacture and apply it. See European Patent Application No. EP1186941A1 and U.S. Patent Application No. 20210349346A1 for detailed manufacturing material and process to make electronically switchable liquid crystal polarizer devices, which are incorporated herein by reference.

The following examples of the present invention show that under three different ambient lighting conditions, the present invention provides improved display ambient contrast ratio (ACR). In the present invention, it is more preferable to have smaller deviation angle to maximize the display ACR performance gain. In these examples, the display is assumed to have a maximum luminance of 800 nits and a native contrast ratio of 1000:1.

Example 1. Overcast Day Ambient Condition

FIG. 9A shows the display ACR dependency on the light incident angle in the case of the ambient illuminance 2000 lux (roughly a typical overcast day ambient condition). The dashed curve 911 is the condition that the incident light onto the display surface is an unpolarized light that has both P-polarization and S-polarization and no polarization filtering control.

The solid curve 912 shows the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction is parallel to P-polarization so that it has 100% P-polarization and 0% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction parallel to the display surface

plane so that the ambient light S-polarization is filtered out but only P-polarization passes through.

The dotted curve **913** shows the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction deviates from P-polarization by 5° so that it has 99% P-polarization and 1% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction orienting at 5° from the display surface plane.

The circle-symbol curve **914** shows the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction deviates from P-polarization by 10° so that it has 97% P-polarization and 3% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction orienting at 10° from the display surface plane.

The square-symbol curve **915** shows the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction deviates from P-polarization by 20° so that it has 88% P-polarization and 12% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction orienting at 20° from the display surface plane.

The triangle-symbol curve **916** is the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction deviates from P-polarization by 30° so that it has 75% P-polarization and 25% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction orienting at 30° from the display surface plane.

As shown in FIG. 9A and Table 1 as follows, all conditions with the linearly polarized incident light having less than 30° deviation angle from P-polarization (the curve **912** through curve **916**) show improved display ACR compared to the unpolarized incident light condition (curve **911**). As the deviation angle between the linear polarization direction and the P-polarization increase from 0° (curve **912**) to 30° (curve **916**), the linearly polarized light has less and less P-polarization composition but more and more S-polarization composition, resulting in gradually reduced display ACR performance gain.

In conclusion, the present invention has no more than 30° deviation angle between the incident light linear polarization and display incident light P-polarization. To obtain good display readability at ambient light conditions, the linear polarization reflecting or absorbing direction of the polarization means is preferably oriented at no more than 30° from the display surface plane.

TABLE 1

Curve	Light polarization	Polarization composition	The angle between the filtering polarizer's reflecting or absorbing direction and the display surface plane
Curve 911	unpolarized	50% P-polarization, 50% S-polarization	n/a
Curve 912	linearly polarized	100% P-polarization, 0% S-polarization	0 DEG
Curve 913	linearly polarized	99% P-polarization, 1% S-polarization	5 DEG
Curve 914	linearly polarized	97% P-polarization, 3% S-polarization	10 DEG

TABLE 1-continued

Curve	Light polarization	Polarization composition	The angle between the filtering polarizer's reflecting or absorbing direction and the display surface plane
Curve 915	linearly polarized	88% P-polarization, 12% S-polarization	20 DEG
Curve 916	linearly polarized	75% P-polarization, 25% S-polarization	30 DEG

Example 2. Full Daylight Ambient Condition

FIG. 9B shows the display ACR dependency on the light incident angle in the case of the ambient illuminance 20000 lux (roughly a typical full daylight ambient condition). The dashed curve **921** is the condition that the incident light onto the display surface is an unpolarized light that has both P-polarization and S-polarization and no polarization filtering control.

The solid curve **922** is the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction is parallel to P-polarization so that it has 100% P-polarization and 0% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction parallel to the display surface plane so that the ambient light S-polarization is filtered out but only P-polarization passes through.

The dotted curve **923** is the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction deviates from P-polarization by 5° so that it has 99% P-polarization and 1% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction orienting at 5° from the display surface plane.

The circle-symbol curve **924** is the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction deviates from P-polarization by 10° so that it has 97% P-polarization and 3% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction orienting at 10° from the display surface plane.

The square-symbol curve **925** is the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction deviates from P-polarization by 20° so that it has 88% P-polarization and 12% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction orienting at 20° from the display surface plane.

The triangle-symbol curve **926** is the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction deviates from P-polarization by 30° so that it has 75% P-polarization and 25% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction orienting at 30° from the display surface plane.

As shown in FIG. 9B and Table 2, all conditions with the linearly polarized incident light having less than 30° deviation angle from P-polarization (the curve **922** through curve **926**) show improved display ACR compared to the unpolarized incident light condition (curve **921**).

larized incident light condition (curve 921). As the deviation angle between the linear polarization direction and the P-polarization increases from 0° (curve 922) to 30° (curve 926), the linearly polarized light has less and less P-polarization composition but more and more S-polarization composition, resulting in gradually reduced display ACR performance gain.

In conclusion, the present invention provides that it is preferable to have no more than 30° deviation angle between the incident light linear polarization and display incident light P-polarization. To obtain good display readability at ambient light conditions, the linear polarization reflecting or absorbing direction of the polarization means is preferably oriented at no more than 30° from the display surface plane.

TABLE 2

Curve	Light polarization	Polarization composition	The angle between the filtering polarizer's reflecting or absorbing direction and the display surface plane
Curve 921	unpolarized	50% P-polarization, 50% S-polarization	n/a
Curve 922	linearly polarized	100% P-polarization, 0% S-polarization	0 DEG
Curve 923	linearly polarized	99% P-polarization, 1% S-polarization	5 DEG
Curve 924	linearly polarized	97% P-polarization, 3% S-polarization	10 DEG
Curve 925	linearly polarized	88% P-polarization, 12% S-polarization	20 DEG
Curve 926	linearly polarized	75% P-polarization, 25% S-polarization	30 DEG

Example 3. Direct Sunlight Ambient Condition

FIG. 9C shows the display ACR dependency on the light incident angle in the case of the ambient illuminance 100000 lux (roughly a typical direct sunlight ambient condition).

The dashed curve 931 is the condition that the incident light onto the display surface is an unpolarized light that has both P-polarization and S-polarization and no polarization filtering control.

The solid curve 932 is the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction is parallel to P-polarization so that it has 100% P-polarization and 0% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction parallel to the display surface plane so that the ambient light S-polarization is filtered out but only P-polarization passes through.

The dotted curve 933 is the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction deviates from P-polarization by 5° so that it has 99% P-polarization and 1% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction orienting at 5° from the display surface plane.

The circle-symbol curve 934 is the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction deviates from P-polarization by 10° so that it has 97% P-polarization and 3% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear

polarization reflecting or absorbing direction orienting at 10° from the display surface plane.

The square-symbol curve 935 is the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction deviates from P-polarization by 20° so that it has 88% P-polarization and 12% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction orienting at 20° from the display surface plane.

The triangle-symbol curve 936 is the condition that the incident light onto the display surface is a linearly polarized light, whose polarization direction deviates from P-polarization by 30° so that it has 75% P-polarization and 25% S-polarization. The condition is achieved by filtering the ambient light with a linear polarizer layer that has the linear polarization reflecting or absorbing direction orienting at 30° from the display surface plane.

As shown in FIG. 9C and Table 3, the present invention provides that all conditions with the linearly polarized incident light having less than 30° deviation angle from P-polarization (the curve 932 through curve 936) show improved display ACR compared to the unpolarized incident light condition (curve 931). As the deviation angle between the linear polarization direction and the P-polarization increases from 0° (curve 932) to 30° (curve 936), the linearly polarized light has less and less P-polarization composition but more and more S-polarization composition, resulting in gradually reduced display ACR performance gain.

In conclusion, the present invention provides that it is preferable to have no more than 30° deviation angle between the incident light linear polarization and display incident light P-polarization. To obtain good display readability at ambient light conditions, the linear polarization reflecting or absorbing direction of the polarization means is preferably oriented at no more than 30° from the display surface plane.

TABLE 3

Curve	Light polarization	Polarization composition	The angle between the filtering polarizer's reflecting or absorbing direction and the display surface plane
Curve 931	unpolarized	50% P-polarization, 50% S-polarization	n/a
Curve 932	linearly polarized	100% P-polarization, 0% S-polarization	0 DEG
Curve 933	linearly polarized	99% P-polarization, 1% S-polarization	5 DEG
Curve 934	linearly polarized	97% P-polarization, 3% S-polarization	10 DEG
Curve 935	linearly polarized	88% P-polarization, 12% S-polarization	20 DEG
Curve 936	linearly polarized	75% P-polarization, 25% S-polarization	30 DEG

I claim:

1. A method for improving readability of a display inside a motorized object, comprising applying a polarization means on a side window of a motorized object, wherein an ambient light propagates through the side window to become a display incident light, the display incident light is reflected by a surface of a display inside the motorized object to form a reflected light, and the display incident light and the reflected light form a display incident plane, and substantially filtering out S-polarization relative to the display incident plane from the ambient light and

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- allowing P-polarization relative to the display incident plane from the ambient light to pass through the side window by the polarization means,
- wherein the polarization means has a linear polarization reflecting or absorbing direction that is substantially parallel to a plane of the surface of the display;
- the polarization means is on the side window that is on a passenger side, a driver side, or both the passenger and driver sides of the motorized object; and
- the side window is a front side window, a rear side window, or both front and rear side windows.
2. The method of claim 1, wherein the polarization means is passively applied on the side window.
3. The method of claim 1, wherein the polarization means is switchable on the side window.
4. The method of claim 1, wherein the side window is of a round, oval, trapezoid, rectangle, square, triangle, or irregular shape.
5. The method of claim 1, wherein the linear polarization reflecting or absorbing direction has no more than 20 degree deviation from being parallel to the plane of the surface of the display.
6. The method of claim 1, wherein the linear polarization reflecting or absorbing direction has no more than 10 degree deviation from being parallel to the plane of the surface of the display.
7. The method of claim 1, wherein the linear polarization reflecting or absorbing direction has no more than 5 degree deviation from being parallel to the plane of the surface of the display.
8. The method of claim 1, wherein the polarization means filters out at least 75% of the S-polarization from the ambient light.

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9. The method of claim 1, wherein the polarization means filters out at least 88% of the S-polarization from the ambient light.
10. The method of claim 1, wherein the polarization means filters out at least 97% of the S-polarization from the ambient light.
11. The method of claim 1, wherein the polarization means filters out at least 99% of the S-polarization from the ambient light.
12. A display readability enhancing system for use in the method of claim 1, comprising the polarization means,
- wherein the polarization means has the linear polarization reflecting or absorbing direction that is substantially parallel to the plane of the surface of the display; and
- the polarization means is on the side window that is on the passenger side, the driver side, or both the passenger and driver sides of the motorized object.
13. The display readability enhancing system of claim 12, wherein the polarization means is located on an interior side of the side window.
14. The display readability enhancing system of claim 12, wherein the polarization means is located on an exterior side of the side window.
15. The display readability enhancing system of claim 12, wherein the polarization means is sandwiched between two layers of the side window.
16. The display readability enhancing system of claim 12, wherein the polarization means is a passive means.
17. The display readability enhancing system of claim 12, wherein the polarization means is switchable.
18. The display readability enhancing system of claim 12, wherein the polarization means filters out at least 75% of the S-polarization from the ambient light.

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